

**FINAL EXAMINATION**  
**STRATEGIC COST ANALYSIS**

**Instructions**

1. **THIS EXAM IS INTENDED TO MEASURE INDIVIDUAL PERFORMANCE. THE CODEs OF CONDUCT OF THE OLIN BUSINESS SCHOOL APPLIES TO THIS EXAM.**
2. The exam consists of 9 pages including this top page. Please, make sure that you have the complete exam.
3. The exam contains four questions. Each question is equally weighted. You should plan on spending about 4-5 hours to complete this exam.
4. The due date of the exam is **11:59pm March 3, 2025**. Please submit the exam on the Canvas course folder under the “Assignment” tab. **You will lose 5 points for each extra day you take for the exam after the due date.** Please plan accordingly.
5. Show all supporting calculations and clearly state any assumption that you make to obtain partial credits in case of a computational error.
6. Feel free to email me (guptam@wustl.edu) if you have any question on the exam. Please note that you should not be talking with anyone other than me to answer any question related to the exam.

**GOOD LUCK!**

## Question 1      **Mason Company**

Bill Alexander is president of the Mason Company, a producer of small size solar panels in a highly competitive market. A recent drop in the price of the solar panels has caused him great concern, since the price is now below the Mason Company's standard cost. Standard cost was determined from operating at 80% of the maximum capacity of 10,000 panels per month. The company does not normally operate above this level (of 8,000 panels per month), since the higher production requires overtime work that significantly increases variable costs.

The fixed costs of the Mason Company are \$40,000 per month for production levels up to 10,000 panels. The variable costs are \$15 per panel for production levels up to 8,000 panels per month. Consequently, the standard cost of the panel is set at \$20, based on operating at 80% capacity. In recent years, the price of the panel ranges from \$21 to \$23, allowing a small but adequate return on the Mason Company's modest investment in machinery and facilities.

For production above the standard volume, unit variable costs for the additional units increase by

- ❖ 15% above the normal variable cost for volume between 80% and 85% of capacity
- ❖ 20% above the normal variable cost for volume between 85% and 90% of capacity
- ❖ 30% above the normal variable cost for volume between 90% and 100% of capacity

Recently the price of panels has dropped about 10% to \$19 per panel. Bill feels that he is now in a no-win situation, since he is losing money on every panel he is selling. While he sees some opportunities for increasing his sales volume above the current level of 8,000 units per month, he believes this would only make matters worse, since he feels that he is losing money at current volumes and the variable costs on the additional units produced will be even greater.

### **Required:**

1. Comment on Bill's analysis of the price-cost squeeze in which he now finds himself. At what point would you recommend that he actually turn down orders at \$19 per panel?

Bill's analysis that his current business plan is not working is a correct one.

Using the equation 1 below, we can calculate a break-even volume number using the new price of \$19. First, I tested the assumption that the standard capacity would not be profitable at this price using information given and equation 1 below.

$$\text{BEP } N^* = \text{Fixed cost} / (\text{Sales price} - \text{Variable cost}) = \text{Fixed Cost} / \text{Contribution Margin}$$

### **Equation 1: Break Even point analysis**

Using the information from the case I plugged in \$40,000 for fixed cost, \$19 for sales price and \$15 to use the variable cost at standard capacity. This gives us:

$$N^* = \$40,000 / (\$19 - \$15) = 10,000 \text{ units}$$

This equation shows us that at standard volume costs, our break even unit number is greater than the standard volumes. Next we construct the full B\* equation that includes overtime variable cost increases to find our true break even unit count. In order to get the N\* number including overtime costs we first use the profit equation below and set it equal to 0:

$$\text{Profit} = 0 = N^* \times \text{price} - (\text{variable cost} \times N^*) - \text{fixed costs}$$

**Equation 2: BEP profit analysis**

$$0 = 19N - (.8 \times 15N + .05 \times 17.25N + .05 \times 18N + .1 \times 19.5N) - 40000$$

$$40000 = 19N - 12N - .8625N - .9N - 1.95N$$

$$40000 = 3.2875N$$

$$N = 12,167.3004 \rightarrow N^* = 12,168 \text{ units}$$

The calculation above shows us that Bill should not do business at volumes below 12,168 units for the price of \$19. This poses a problem to Bill's company as their capacity is currently only 10k. In the current circumstances, Bill is right. He is in a lose-lose and his current business cannot make money without either, reducing costs, increasing price, or increasing capacities which would increase fixed costs

2. Assuming the price increases to \$19.75, at what volumes will the Mason Company operate profitably?

$$0 = 19.75N - (.8 \times 15N + .05 \times 17.25N + .05 \times 18N + .1 \times 19.5N) - 40000$$

$$40000 = 19.75N - 12N - .8625N - .9N - 1.95N$$

$$40000 = 4.0375N$$

$$N^* = 9,907.12 \rightarrow N^* = 9,908 \text{ units}$$

## Question 2      Hammer Products, Inc.

“Wow! Is that B-10 model ever a loser! I say the time has come cut back its production and shift our resources toward the new C-20 model,” said Rory Moncur, executive vice president of Hammer Products, Inc. “Just look at this statement I’ve received from accounting. The C-20 is generating twice as much in profits as the B-10, and it has only about one-fifth as much in sales. I’m convinced that our future depends on the C-20.” The year-end statement to which Rory was referring follows:

| HAMMER PRODUCTS, INC.                          |                   |                   |                   |
|------------------------------------------------|-------------------|-------------------|-------------------|
| Income Statement                               |                   |                   |                   |
|                                                |                   | Product Model     |                   |
|                                                | Total             | B-10              | C-20              |
| Sales .....                                    | \$14,500,000      | \$12,000,000      | \$2,500,000       |
| Cost of goods sold.....                        | <u>9,000,000</u>  | <u>7,200,000</u>  | <u>1,800,000</u>  |
| Gross margin .....                             | 5,500,000         | 4,800,000         | 700,000           |
| Less selling and administrative expenses ..... | <u>4,900,000</u>  | <u>4,600,000</u>  | <u>300,000</u>    |
| Net income .....                               | <u>\$ 600,000</u> | <u>\$ 200,000</u> | <u>\$ 400,000</u> |
| Number of units produced and sold .....        | —                 | 60,000            | 10,000            |
| Net income per unit.....                       | —                 | \$3.33            | \$40.00           |

“The numbers sure look that way,” replied Connie Collins, the company’s sales manager. “But why isn’t the competition more excited about the C-20? I know we’ve only been producing the model for three years, but I’m surprised that more of our competitors haven’t recognized what a cash cow it is.”

“I think it’s our new automated plant,” replied Rory. “Now it takes only one direct labor-hour to produce a unit of the B-10 and one-and-a-half direct labor-hours to produce a unit of the C-20. That’s considerably less than it used to take us.”

“I agree that automation is wonderful,” replied Connie. “I suppose that’s how we’re able to hold down the price of the C-20. Borst Company in Germany tried to bring out a C-20 but discovered that they couldn’t touch our price. But Borst is killing us on the B-10 by undercutting our price with some of our best customers. I suppose they’ll pick up all of our B-10 business if we move out of that market. But who cares? We don’t even have to advertise the C-20; it just seems to sell itself.”

“My only concern about automation is how our manufacturing overhead rate has shot up,” said Rory. “Our total manufacturing overhead cost is \$3,600,000. That comes out to be a hefty amount per direct labor-hour, but old Fred down in accounting has been using labor-hours as the

base for computing overhead rates for years and doesn't want to change. I don't suppose it matters so long as costs get assigned to products."

"I've never understood that debit and credit stuff," replied Connie. "But I think you've got a problem in production. I had lunch with Joanne yesterday and she complained about how complex the C-20 is to produce. Apparently, they have to do a lot of setups, special soldering, and other work on the C-20 just to keep production moving. And they have to inspect every single unit.

"That all have to wait," said Rory. "I'm writing a proposal to the board of directors to phase out the B-10. We've got to bring our bottom line up or we'll all be looking for jobs."

**Required:**

1. Compute the current budgeted overhead rate per direct labor hour. (You may assume that there was no under- or over-applied overhead for the year.)

I calculated the total hours of labor for each model during the year:

Hours Labor B-10 = 1 hour per unit \* 60,000 units = 60,000 hours

Hours Labor C-20 = 1.5 hours per unit \* 10,000 units = 15,000 hours.

Total hours labor = 60,000 + 15,000 = 75,000 hours

Total overhead cost: \$3,600,000 / 75,000 hours = \$48 per labor hour.

2. Materials and labor costs per unit for the two products are as follows:

|                       | <b>B-10</b> | <b>C-20</b> |
|-----------------------|-------------|-------------|
| Direct materials..... | \$60        | \$90        |
| Direct labor .....    | 12          | 18          |

Using these data and the rate computed in (1) above, determine the unit product cost of each product.

Using the OVHD cost per labor hour from table 1 we calculate the model product cost below:

B-10 Product Cost = 60 + 12 + 1\*48= \$120

C-20 Product Cost = 90 + 18+ 1.5\* 48 = \$180

3. Assume that the company's \$3,600,000 in manufacturing overhead cost is traceable to six activity centers, as follows:

| Activity Center (and Cost Driver)         | Estimated Overhead Costs  | Expected Activity-Cost Driver Units |         |         |
|-------------------------------------------|---------------------------|-------------------------------------|---------|---------|
|                                           |                           | Total                               | B-10    | C-20    |
| Machine setups (number of setups) .....   | \$416,000                 | 3,200                               | 2,000   | 1,200   |
| Quality control (number of inspections)   | \$720,000                 | 18,000                              | 8,000   | 10,000  |
| Purchase orders (number of orders).....   | \$180,000                 | 2,400                               | 1,680   | 720     |
| Soldering (number of solder points) ..... | \$900,000                 | 400,000                             | 120,000 | 280,000 |
| Shipments (number of shipments) .....     | \$264,000                 | 1,200                               | 800     | 400     |
| Machine related (machine-hours) .....     | <u>\$1,120,000</u>        | 140,000                             | 60,000  | 80,000  |
| <b>Total</b>                              | <b><u>\$3,600,000</u></b> |                                     |         |         |

Given these data, compute (1) the cost allocation rate at each activity center; and (2) the ABC unit product cost each product.

Sell through volume B-10 = 60,000

Sell through volume C-20 = 10,000

- Machine setup allocation =  $\$416,000 / 3200 = \$1,300$  per setup
  - o Setups per unit B-10 =  $2,000 / 60,000 = 0.033$  setups per unit
  - o Setups per unit C-20 =  $1,200 / 10,000 = 0.12$  setups per unit
  - o Setup allocation per unit B-10 =  $\$1,300 * 0.033 = \$43.33$  per unit
  - o Setup allocation per unit C-20 =  $\$1,300 * 0.12 = \$156$  per unit
- Quality Control allocation =  $\$720,000 / 18,000 = \$40$  per inspection
  - o Inspections per unit B-10 =  $8,000 / 60,000 = 0.133$  inspections per unit
  - o Inspections per unit C-20 =  $10,000 / 10,000 = 1$  inspection per unit
  - o QC allocation per unit B-10 =  $\$40 * 0.133 = \$5.32$  per unit
  - o QC allocation per unit C-20 =  $\$40 * 1 = \$40$  per unit
- Purchase order allocation =  $\$180,000 / 2,400 = \$75$  per PO
  - o PO's per unit B-10 =  $1,680 / 60,000 = 0.028$  PO's per unit
  - o PO's per unit C-20 =  $720 / 10,000 = 0.072$  PO's per unit
  - o Order allocation per unit B-10 =  $\$75 * 0.028 = \$2.10$  per unit
  - o Order allocation per unit C-20 =  $\$75 * 0.072 = \$5.40$  per unit
- Soldering allocation =  $\$900,000 / 400,000 = \$2.25$  per solder point
  - o solders per unit B-10 =  $120,000 / 60,000 = 12$  solders per unit
  - o solders per unit C-20 =  $280,000 / 10,000 = 28$  solders per unit
  - o solder allocation per unit B-10 =  $\$2.25 * 12 = \$27$  per unit
  - o solder allocation per unit C-20 =  $\$2.25 * 28 = \$63$  per unit

- Shipment allocation=  $\$264,000 / 1,200 = \$220$  per shipment
  - o Shipments per unit B-10  $800 / 60,000 = 0.013$  shipments per unit
  - o Shipments per unit C-20  $= 400 / 10,000 = 0.04$  shipments per unit
  - o shipping allocation per unit B-10  $= \$220 * 0.013 = \$2.86$  per unit
  - o shipping allocation per unit C-20  $= \$220 * 0.04 = \$8.80$  per unit
- Machine related allocation=  $\$1,120,000 / 140,000 = \$8$  per machine hour
  - o M-hours per unit B-10  $60,000 / 60,000 = 1$  M-hours per unit
  - o M-hours per unit C-20  $= 80,000 / 10,000 = 8$  M-hours per unit
  - o M-hour allocation per unit B-10  $= \$8 * 1 = \$8$  per unit
  - o M-hour allocation per unit C-20  $= \$8 * 8 = \$64$  per unit

B-10 Overhead cost per unit =  $43.33 + 5.32 + 2.10 + 27 + 2.86 + 8 = \$88.61$

C-20 Overhead cost per unit =  $156 + 40 + 5.40 + 63 + 8.80 + 64 = \$337.20$

B-10 ABC unit product cost =

4. Would you support Rory's recommendation to expand sales of the C-20? Explain your analysis to help the board of directors make a decision. Using the info from the ABC activity above I would absolutely not recommend expanding sales of the C-20. The overhead costs associated with producing that product are far greater than the costs of producing B-10. The manufacturing group is spending heavily to run their manufacturing line for that product and the cost is hidden when using cost by direct labor hours to allocate the overhead costs. The product has already been produced for three years and the costs to produce C-20 are still far higher than B-10.
5. If you were president of Hammer Products, Inc., what strategy would you follow to improve the company's overall profits?

For the reasons above, I would recommend reducing the cost of B-10 to better compete in that market where they are able to make more money. They need to increase the price of C-20 to offset the additional costs to manufacture or get out of the C-20 market.

### Question 3.      Sulu Enterprise

The Red Phaser division of Sulu Enterprises expects to report a pretty good year to headquarters and to shareholders. They expect to produce and sell 40,000 phasers throughout the galaxy this year. While the machinery currently used in the production process is fairly old, it is not very expensive to operate. It has a current book value of \$250,000, one remaining year of useful life, and no expected salvage value at the end of the year. However, the machine could be sold for \$100,000 now.

Because the machinery is not state-of-the-art, the production process is fairly labor intensive. As a result, the company's labor costs are high. In addition, the current production process also consumes a lot of variable overhead.

A factor that is expected to contribute to this year's profitability is the foresight (or luck) of Red Phaser's purchasing manager, Mr. Rickey. Mr. Rickey purchased 40,000 pounds of Alpha, the raw material that the current machinery uses to produce phasers, at the end of last year for \$10 per pound. This was quite fortunate, since the price of Alpha has subsequently skyrocketed, and now costs \$20 per pound. Under current operating conditions, the 40,000 pounds of Alpha on hand is enough to satisfy this year's production of 40,000 units of output (each unit of output requires 1 pound of raw material Alpha on the currently used machine). Another factor that is expected to add to this year's success is a current change in the policy of the federation of republics in the galaxy. The new policy has banished taxes on corporate income. Using the current machine, the company therefore expects the costs that will appear in their operating income for the coming year to be

|                            |                               |                |
|----------------------------|-------------------------------|----------------|
| Direct Materials           | 1 lb. of Alpha @ \$10 per lb. | \$10.00        |
| Direct Labor               | 1.6 hrs. @ \$12.50 per hr.    | 20.00          |
| Variable Overhead          | @ 40% of Direct Labor         | 8.00           |
| Fixed Overhead             | \$250,000/40,000 units        | 6.25           |
| <b>Total Cost Per Unit</b> |                               | <b>\$44.25</b> |

The manager of the Red Phaser division, Mr. Spark, is considering replacing the old machine with a new one. The new machine will offer the flexibility of being capable of processing either Raw Material Alpha or a newly developed synthetic raw material Beta. However, the new machine does not work very efficiently with raw material Alpha. Half of the Alpha gets wasted in the process, which means that 2 lbs. of raw material Alpha are necessary to produce a unit of output with the new machine. The new machine is very efficient at using the new material — it requires 1 pound of the synthetic material Beta to produce a unit of output. Neither the choice of the machine nor the choice of the raw material is expected to have any perceptible effect on the quality of the finished product. Production and sales volumes are expected to be 40,000 units regardless of which machine is used.

The new machine would be rented for \$300,000. While it is expensive, the new machine

production process is also more efficient than the old machine. As a result, the labor time would be cut by 25 percent. Dr. Scotty, the production engineer of Red Phaser, favors replacing the old machine. He argues that if the new machine is obtained, raw material Beta should be used because it is less expensive than either the old Material Alpha or any new Alpha that might have to be purchased. That is, he calculates that it will cost \$12 for materials to make a unit of output using raw material Beta (i.e., 1 lb. of Beta @ \$12/lb.), versus \$20 of Alpha (at the new price of Alpha) on the old machine. Mr. Rickey informed Dr. Scotty that, if not used, the existing stock of Alpha can be sold in the market, at a price of \$12 per pound.

To support his argument for acquiring the new machine, Dr. Scotty provides the following calculation of Red Phaser's expected operating costs with the new machine.

|                            |                              |                |
|----------------------------|------------------------------|----------------|
| Direct Materials           | 1 lb. of Beta @ 12 per lb.   | \$12.00        |
| Direct Labor               | 1.2 hours @ \$12.50 per hour | 15.00          |
| Variable Overhead          | @ 40% of Direct Labor        | 6.00           |
| Fixed Overhead             | \$300,000/40,000 units       | 7.50           |
| <b>TOTAL COST PER UNIT</b> |                              | <b>\$40.50</b> |

**Required:**

1. What should Red Phaser do? Support your answer with relevant cost analysis and a discussion of all relevant factors.

The first issue to resolve with this question is which problem do we really need to solve? Given the new price of Alpha and estimated costs to manufacture using alpha or beta, we know that ultimately Red Phaser will need to transition their processes to use Beta. The question we need to resolve is what should we do with the remaining inventory of Alpha? We have three options, should we a) resell the Alpha on hand b) manufacture it on the old machine c) manufacture it on the new machine and sell the old machine.

b) costs of manufacturing alpha on the old machine:

|                            |                               |                |
|----------------------------|-------------------------------|----------------|
| Direct Materials           | 1 lb. of Alpha @ \$10 per lb. | \$10.00        |
| Direct Labor               | 1.6 hrs. @ \$12.50 per hr.    | 20.00          |
| Variable Overhead          | @ 40% of Direct Labor         | 8.00           |
| Fixed Overhead             | \$250,000/40,000 units        | 6.25           |
| <b>Total Cost Per Unit</b> |                               | <b>\$44.25</b> |

The cost of manufacturing using the alpha product on the old machine are given as \$44.25 per unit.  $\$44.25 \times 40,000 \text{ units} = \$1,770,000$ . This will exhaust the value of our old machine as well with adds + \$100,000 in opportunity cost. We add a second missed

opportunity cost for the resale value of the Alpha material  $40,000 \text{ lbs} * \$12$  gives \$480,000 additional. Our total cost is  $\$1,770,000 + \$100,000 + \$480,000 = \$2,350,000$ .

c) If we choose to manufacture the alpha material on the new machine and resell the old machine we use the two cost tables below.

|                            |                              |                |
|----------------------------|------------------------------|----------------|
| Direct Materials           | 2 lb. of Alpha @ 10 per lb.  | \$20.00        |
| Direct Labor               | 1.2 hours @ \$12.50 per hour | 15.00          |
| Variable Overhead          | @ 40% of Direct Labor        | 6.00           |
| Fixed Overhead             | \$300,000/40,000 units       | 7.50           |
| <b>TOTAL COST PER UNIT</b> |                              | <b>\$48.50</b> |
|                            |                              |                |
| Direct Materials           | 1 lb. of Beta @ 12 per lb.   | \$12.00        |
| Direct Labor               | 1.2 hours @ \$12.50 per hour | 15.00          |
| Variable Overhead          | @ 40% of Direct Labor        | 6.00           |
| Fixed Overhead             | \$300,000/40,000 units       | 7.50           |
| <b>TOTAL COST PER UNIT</b> |                              | <b>\$40.50</b> |

We would then produce 20k units using the alpha material at a cost of  $\$48.50 * 20k = \$970,000$ , resell the old machine for a profit of 100k, then manufacture 20k units using Beta at the cost  $\$40.50 * 20k = \$810,000$ . We add the cost of missed opportunity to resell alpha material  $\$12 * 40,000 = \$480,000$ . We calculate the total cost using the equation:  $\$970,000 + \$810,000 + \$480,000 - \$100,000 = \$2,160,000$ .

- a) Resell the alpha product and machine and produce all 40k units of Beta on the new machine. First, we get the cost of producing 40k units of beta on the new machine:  $\$40.50 * 40,000 = \$1,620,000$ . We subtract the resale value of 40k lbs of alpha  $40,000 * \$12 = \$480,000$ . We subtract the value of selling the old machine \$100,000. This gives us a total cost  $\$1,620,000 - \$480,000 - \$100,000 = \$1,040,000$ .

Red Phaser should proceed with option A. They should resell the alpha product on hand and the old machine. They should begin producing immediately on a new machine using Beta material.

2. Ignore your answer to part (1). The comparison in the statement of the problem implicitly assumes that direct labor costs change depending on which machine is used. That is, the new machine will decrease labor costs by 25 percent. According to the galactic labor laws, the company must pay 75% of wages to any displaced workers for one year. However, the displaced workers can be redeployed for some other less productive activities without having to pay full 100% of wages. How does this change your analysis, if

- company has no other activities to engage the displaced workers? Or,
- have some less productive activities to use the displaced workers?

|                            |                              |                |
|----------------------------|------------------------------|----------------|
| Direct Materials           | 1 lb. of Beta @ 12 per lb.   | \$12.00        |
| Direct Labor               | 1.2 hours @ \$12.50 per hour | 21.75          |
| Variable Overhead          | @ 40% of Direct Labor        | 6.00           |
| Fixed Overhead             | \$300,000/40,000 units       | 7.50           |
| <b>TOTAL COST PER UNIT</b> |                              | <b>\$47.25</b> |

The table above takes the beta manufacturing cost table and adds back in 75% of the direct labor cost reduction. First, we get the cost of producing 40k units of beta on the new machine:  $\$47.25 \times 40,000 = \$1,890,000$ . We subtract the resale value of 40k lbs of alpha  $40,000 \times \$12 = \$480,000$ . We subtract the value of selling the old machine \$100,000. This gives us a total cost  $\$1,890,000 - \$480,000 - \$100,000 = \$1,310,000$ . Regardless of the extra tasks these laborers can be assigned too, plan a is still the best plan. Our answer does not change.

Does the labor law favors replacing the old machine or keeping it? Support your answer with relevant cost analysis and a discussion of all relevant factors.

The labor laws ultimately favor replacing the machine in my mind. They don't make much of a difference in our analysis, but they do add some cost to the current fiscal year. Selling the machine while it still has value and selling off the alpha product give us some cash income to use to pay those expenses.

3. Suppose in addition to information in question 2, the sales volume for Red Phaser is expected to be considerably higher than 40,000, will this favor replacing or keeping the old machine?

An increase in volume favors keeping the old machine to manufacture all the alpha material into product. There is more value in selling product manufactured from alpha than there is in selling the alpha material and machine directly, but the most value comes from selling product manufactured from material beta. If our sales volume is not a limiting factor and we are not at our production capacity, it would be the most valuable to manufacture product on the old machine using alpha material until our alpha material is used up, then begin producing product on a new machine with beta product.

## Question 4.      Molina Company

“It certainly is nice to see that small variance on the income statement after all the trouble we’ve had lately in controlling manufacturing costs,” said Linda White, vice president of Molina Company. The \$2,250 variance reported last period is well below the 3% limit we have set for variances. We need to congratulate everybody on a job well done.” The income statement to which Ms. White was referring is shown below:

### 20,000 Units

|                                     | <b>Actual</b>    | <b>Expected*</b> |
|-------------------------------------|------------------|------------------|
| Sales .....                         | \$1,200,000      | \$1,200,000      |
| Less cost of goods sold             |                  |                  |
| (standard cost, \$38 per unit)..... | <u>762,250</u>   | <u>760,000</u>   |
| Gross margin .....                  | <u>437,750</u>   | <u>440,000</u>   |
| Less operating expenses:            |                  |                  |
| Selling expenses.....               | 200,000          | 200,000          |
| Administrative expenses .....       | <u>150,000</u>   | <u>150,000</u>   |
| Total operating expenses.....       | <u>350,000</u>   | <u>350,000</u>   |
| Net income .....                    | <u>\$ 87,750</u> | <u>\$ 90,000</u> |

\* Expected COGS is based on 20,000 units produced and sold using standard cost rates given below.

The company produces and sells a single product. The standard cost card for the product are based on expected labor usage of 25,000 hours for the year and the budgeted fixed manufacturing overhead of \$150,000.

### Standard Cost Card – Per Unit

|                                           |      |
|-------------------------------------------|------|
| Direct materials, 4 yards at \$3.50 ..... | \$14 |
| Direct labor, 1.5 hours at \$8.....       | 12   |
| Variable overhead, 1.5 hours at \$2 ..... | 3    |
| Fixed overhead, 1.5 hours at \$6 .....    | 9    |
| Standard cost per unit.....               | \$38 |

The following additional information is available for the year just completed:

- a. The company manufactured and sold 20,000 units of product during the year.
- b. A total of 78,000 yards of material were purchased during the year at a cost of \$3.75 per yard. All of this material was used to manufacture the 20,000 units. There were no beginning or ending inventories for the year.
- c. The company worked 32,500 direct labor-hours during the year at a cost of \$7.80 per hour.
- d. Overhead cost is applied to products on the basis of direct - labor-hours. Data relating to overhead costs follow:

|                                |           |
|--------------------------------|-----------|
| Actual fixed overhead costs    | \$148,000 |
| Actual variable overhead costs | \$68,250  |

**Required:**

1. Analyze variances as best as you can for the year. *Hint: you will need to calculate:*
  - a. Material price variance.
    - $78,000 \text{ yards} * \$ 3.75 = \$292,500 / 20,000 \text{ units} = \$14.625 \text{ per unit}$
    - $\text{Material price variance} = \$14 - \$14.625 = (\$0.625)$
  - b. Material usage variance.
    - $78,000 \text{ yards} / 20,000 \text{ units} = 3.9 \text{ yards per unit}$
    - $\text{Material usage variance} = 4 \text{ yards} - 3.9 \text{ yards} = 0.1 \text{ yards}$
  - c. Labor efficiency variance.
    - $32,500 \text{ hours} / 20,000 \text{ units} = 1.625 \text{ hours per unit}$
    - $\text{Labor efficiency variance} = 1.5 \text{ hours per unit} - 1.625 \text{ hours per unit} = (0.125)$
  - d. Labor price variance, and
    - $32,500 * \$7.80 = \$253,500 / 20,000 = \$12.675 \text{ direct labor dollars per unit}$
    - $\text{Labor price variance} = \$12 - \$12.675 = (0.675)$
  - e. Var. overhead. efficiency variance.
    - $\$68,250 / 32,500 = \$2.10 \text{ per hour}$
    - $\text{Var. overhead efficiency variance} = \$2 - \$2.10 = (\$0.10)$
  - f. Var. overhead spending variance, and
    - $\$68,250 / 20,000 \text{ units} = \$3.4125$
    - $\text{Variable overhead efficiency variance} = \$3 - \$3.4125 = (\$0.4125)$
  - g. Fixed overhead spending variance
    - $\$148,000 / 20,000 \text{ units} = \$7.40 \text{ per unit}$
    - $\text{Fixed overhead spending variance} = \$7 - \$7.40 = (\$0.40)$
  - h. Production volume variance
    - $20,000 \text{ units expected} - 20,000 \text{ units produced} = 0 \text{ production volume variance}$
2. Do you agree that everyone should be congratulated for a job well done? Explain.

From the data collected in problem 1 we are able to correct the actual standard cost table to gather our COGS. That corrected table is below.

**Actual Standard Cost Card – Per Unit**

|                                               |          |
|-----------------------------------------------|----------|
| Direct materials, 3.9 yards at x .....        | \$14.625 |
| Direct labor, 1.65 hours at \$x.....          | 12.675   |
| Variable overhead, 1.65 hours at \$2.10 ..... | 3.465    |
| Fixed overhead, 1.65 hours at \$7.40 .....    | 12.21    |
| Standard cost per unit.....                   | \$42.975 |

The actual standard cost per unit is \$42.975 multiplying that by 20,000 gives us the corrected income table below.

**20,000 Units**

|                                     | <b>Actual</b>    | <b>Expected*</b> |
|-------------------------------------|------------------|------------------|
| Sales .....                         | \$1,200,000      | \$1,200,000      |
| Less cost of goods sold             |                  |                  |
| (standard cost, \$38 per unit)..... | <u>859,500</u>   | <u>760,000</u>   |
| Gross margin .....                  | <u>340,500</u>   | <u>440,000</u>   |
| Less operating expenses:            |                  |                  |
| Selling expenses.....               | 200,000          | 200,000          |
| Administrative expenses .....       | <u>150,000</u>   | <u>150,000</u>   |
| Total operating expenses.....       | <u>350,000</u>   | <u>350,000</u>   |
| Net income                          | <u>\$ -9,500</u> | <u>\$ 90,000</u> |

In actuality our net income is a loss of \$9,500 from our corrected COGS value. Our variance is actually 89%, our income is negative, and COGS were calculated incorrectly. Only our material usage has improved. Really no one on this team should be congratulated for this period of performance.